



AuthenticOne — Project Introduction

AuthenticOne is a AI-diven trading operating system. It watches the market, turns raw data into trade decisions, manages risk, and executes trades with a broker — with minimal manual work from the trader.

It's built as a **modular monolith**: one deployable application made of **9 independent modules**, each owning one clear responsibility, communicating with each other through events rather than reaching into each other's data.

Data Pipeline

The modules that get market data in and turn it into something the rest of the system can use.

- **Market Data & Feature Processing** — Pulls in market data, and computes the analytical output the rest of the system runs on — internally called a **Market Reading** (momentum, volatility, trend, and similar signals for an asset, on a given date).

Intelligence & Decisioning

Two specialized AI agents, each with one job — not one black box trying to do everything.

- **Signal Generation** — Runs the **Decision Agent**: reads a Market Reading and decides whether a valid trade exists, and if so, which direction.
- **Risk Management** — Runs the **Monitoring Agent**: sizes the position, sets stop loss/take profit, and manages the trade live until it closes — all within guardrails the trader controls (max daily/weekly loss, max consecutive losses), and a platform-wide emergency halt.

Trading

- **Execution** — Sends orders to the trader's connected broker and tracks them through to fill. Built to support multiple brokers, though each trader connects just one at a time.

Platform / Supporting Modules

Standard product needs, independent of the trading logic itself.

- **Identity & Access** — Accounts, login (including social login), sessions.
 - **Subscription** — Free vs. subscribed access tiers.
 - **Watchlist / Asset Catalog** — The platform's supported-asset list, and what each trader is watching.
 - **Notification** — Turns events from every other module into alerts a trader actually sees.
 - **Platform Administration** — System health, audit trail, admin oversight.
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Tech Stack

- **Spring Boot** — Application framework
- **PostgreSQL** — Primary database
- **Redis** — Rolling market-data window and caching
- **Kafka** — Event-driven communication between modules
- **GraphQL** — Read API
- **REST** — Command (write) API

Key Integrations

- **Alpaca Markets** — Market data provider
 - **MT5 and Tradovate** — Broker integrations
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This is a high-level map of the system's components — not a description of the algorithms or models inside them.